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Martti Karvonen

Employment

2026-present **Lecturer**, *University of Bath*

2023-2026 **Postdoctoral fellow**, *University College London*

2019-2023 **Postdoctoral fellow**, *University of Ottawa*

Education

2015-2019 **PhD**, *University of Edinburgh*, Supervisors: Chris Heunen and Tom Leinster

2014-2015 **PhD studies**, *Aalto University*

2013-2014 **Master of Science in Mathematics and the Foundations of Computer Science**, *University of Oxford*
Graduated with distinction.

2010-2012 **Bachelor of Science**, *University of Helsinki*
Major: Mathematics. Minor: Economics.

2009-2012 **Bachelor of Arts**, *University of Helsinki*
Major: Theoretical Philosophy. Minor: Philosophy of Values and World Views.

Grants, scholarships and awards

2025 **Seal of Excellence**, *Marie Skłodowska Curie Actions, EU*

2020-2025 **Research grant: Cryptography in a Quantum World**, *US Air Force Office of Scientific Research*, Wrote the section concerning my joint project with the PI, representing approximately 25% of the application text, and 50% of the proposed research, Principal Investigator: Anne Broadbent

2016 **Short-term scientific mission: Reversible monads and reversible effects**, *COST Action IC1405*, STSM Location: University of Copenhagen

2016-2018 **Postgraduate scholarship**, *The Osk.Huttunen foundation*

2015 **Postgraduate scholarship**, *Vilho, Yrjö and Kalle Väisälä Foundation*

2013 **Outstanding mathematics student award**, *The Mathematics and Science Fund, University of Helsinki*

2012 **Outstanding philosophy student award**, *The Ilkka Niiniluoto Fund, University of Helsinki*

Teaching experience

Summer 2021 **Reading Course on Category Theory**, *Lecturer*
University of Ottawa, Department of Mathematics and Statistics

Winter 2021 **Introduction to Mathematical Logic**, *Lecturer*
University of Ottawa, Department of Mathematics and Statistics

Winter 2020 **Introduction to Linear Algebra**, *Lecturer*
University of Ottawa, Department of Mathematics and Statistics

Spring 2019 **Introduction to Theoretical Computer Science**, *Lecturer responsible for half of the course*
University of Edinburgh, School of Informatics.

Spring 2018 **Introduction to Theoretical Computer Science**, *Tutor*
University of Edinburgh, School of Informatics.

Spring 2017 **Algorithms, Data Structures and Learning**, *Tutor*
University of Edinburgh, School of Informatics.

Spring 2016 **Algorithmic Game Theory and Applications**, *Tutor and marker*
University of Edinburgh, School of Informatics.

Spring 2015 **Introduction to discrete mathematics**, *TA and marker*
Aalto University, Department of Mathematics and Systems analysis.

Summer 2013 **Measure and Integral**, *TA*
University of Helsinki, Department of Mathematics and Statistics.

Springs 2012-2013 **Logic: basic part and Logic: supplementary part I**, *TA*
University of Helsinki, Department of Philosophy.

Falls 2010-2012 **Introduction to logic**, *TA*
University of Helsinki, Department of Philosophy.

Supervision

Summer 2026 **Adjoint school**, *Project mentor*
International Applied Category Theory Conference

10/2025- **Andrea Corradetti**, *Research internship*
University College London, Department of Computer Science

10/2025-11/2025 **Lucas Stinchcombe**, *Quantum CDT Case study*
University College London, Department of Computer Science

10/2024-9/2025 **Max Quartly**, *MSc project*
University College London, Department of Computer Science

10/2024-11/2024 **Delia Melinte Citea**, *Quantum CDT Case study*
University College London, Department of Computer Science

10/2023-9/2024 **Kostas Kvietinskas**, *MSc project*
University College London, Department of Computer Science

Summer 2021 **Tanguy Massacrier**, *Research internship*
University of Ottawa, Department of Mathematics and Statistics

Reviewer

Books Springer, Oxford University Press

Journals Journal of Mathematical Physics, Communications in Mathematical Physics, Nature Communications, PRX Quantum, Physical Review Letters, Physical Review A, Journal of Physics A: Mathematical and Theoretical, Quantum, Quantum Information Processing, Reports on Progress in Physics, Annals of Pure and Applied Logic, Journal of Pure and Applied Algebra, Journal of Algebra and Its Applications, Compositionality, Theory and Applications of Categories, Applied Categorical Structures, Journal of Logical and Algebraic Methods in Programming

Conference PC member — Quantum Physics and Logic (QPL) 2022-2025, Applied Category Theory 2021 & 2025, 26th International Workshop on Algebraic Development Techniques (WADT 2022), Fourth Symposium on Compositional Structures (SYCO 4)

Conference subreviewer — Logic in Computer Science, Computer Science Logic, Quantum Information Processing, Quantum Physics and Logic, Eurocrypt, Foundations of Software Science and Computation Structures, Theory of Cryptography Conference

Departmental Service

2021-2023 **Organization of the department's quantum cryptography seminar**, *University of Ottawa, Department of Mathematics and Statistics*

Talks

Selected invited seminar and workshop talks

5/2025 **Mathematical Foundations of Quantum Advantage**
Simon Fraser University

4/2025 **Thirteenth Symposium on Compositional Structures**
University College London

5/2024 **The Oxford Advanced Seminar on Informatic Structures**
University of Oxford

3/2024 **Workshop on Process Theory for Security Protocols and Cryptography**
TallTech

10/2023 **Atlantic Category Theory Seminar**
Dalhousie University

3/2023 **Seminar of Chalmers Security & Privacy Lab**
Chalmers University of Technology

2/2023 **Quantum Cryptography Seminar**
Université de Montréal

1/2023 **AMS Special Session on Applied Category Theory**, *Joint Mathematics Meeting*
Boston

9/2022 **Quantum Cryptography Seminar**
TU Berlin

8/2022 **Bilkent University Quantum Computing Seminar**
Bilkent University

11/2021 **Quantum Seminar**
University of Geneva

10/2021 **Quantum Cryptography Seminar**
ETH Zürich

5/2021 **Category theory seminar**
Cambridge

8/2020 **TallCat seminar**
Tallinn

12/2019 **Logic and Structure in Computer Science and Beyond**
Leiden

7/2019 **Workshop on Contextuality as a resource in quantum computation**
Oxford

3/2019 **The Oxford Advanced Seminar on Informatic Structures**
Oxford

9/2018 **Category Theory Seminar**
Cambridge

3/2018 **Combining Viewpoints in Quantum Theory**
Edinburgh

11/2016 **Categories, Logic, and Physics, Scotland**
Glasgow

10/2016 **Copenhagen Programming Language Seminar**
Copenhagen

Selected contributed talks

7/2022 **17th Conference on the Theory of Quantum Computation, Communication and Cryptography (TQC 2022)**
University of Urbana-Champaign

4/2022 **25th International Conference on Foundations of Software Science and Computation Structures (FoSSaCS 2022)**
Munich

8/2021 **International Category Theory Conference**
Genoa

7/2021 **International Conference on Applied Category Theory**
Cambridge

6/2021 **Structure meets Power Workshop**
LICS 2021 Satellite event

6/2021 **Canadian Mathematical Society summer meeting**
Ottawa

5/2021 **Workshop: Quantum Contextuality in Quantum Mechanics and Beyond**
Prague

7/2019 **Applied Category Theory Conference**
Oxford

7/2019 **International Category Theory Conference**
Edinburgh

5/2019 **Workshop: Quantum Contextuality in Quantum Mechanics and Beyond**
Prague

6/2018 **15th International Conference on Quantum Physics and Logic**
Halifax

4/2018 **103rd Peripatetic Seminar on Sheaves and Logic**
Brno

8/2016 **International Category Theory Conference**
Halifax

6/2015 **Mathematical Foundations of Programming Semantics XXXI**
Nijmegen

Publications

With the exception of [1, 3], all authors are listed in alphabetical order by last name. This is in line with common practice in mathematics and computer science-—see e.g. the culture statement from the American Mathematical Society:
`http://www.ams.org/profession/leaders/culture/CultureStatement04.pdf`

Journal Articles

[1] B. M. Bumpus, W. Leal, J. Fairbanks, M. Karvonen, and F. Simard. Towards a unified theory of time-varying data. *Applied Categorical Structures*, 34(3), 2026.

[2] M. Karvonen. A comment on: 'The axiom of choice and the no-signalling principle' (2025), by Baumeler et al. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 482, 2026. To appear.

[3] F. Wiesner, Z. Chaoui, D. Kessler, A. Pappa, and M. Karvonen. Why cut-and-choose quantum state verification cannot be both efficient and secure. *IACR Communications in Cryptology*, 2(4), 2026.

[4] A. Broadbent, M. Karvonen, and S. Lord. Uncloneable quantum advice. *IACR Communications in Cryptology*, 1(3), 2024.

[5] P. Hofstra and M. Karvonen. Inner autoequivalences in general and those of monoidal categories in particular. *Journal of Pure and Applied Algebra*, 228(11):107717, 2024.

[6] P. Hofstra and M. Karvonen. Inner automorphisms as 2-cells. *Theory and Applications of Categories*, 42(2):19–40, 2024.

[7] A. Broadbent and M. Karvonen. Categorical composable cryptography: extended version. *Logical Methods in Computer Science*, 19:30:1–30:46, 4, 2023.

[8] M. Karvonen. Neither Contextuality nor Nonlocality Admits Catalysts. *Physical Review Letters*, 127:160402, 16, 2021.

[9] M. Karvonen. Biproducts without pointedness. *Cahiers de topologie et géométrie différentielle catégoriques*, 61(3):229–238, 2020.

[10] C. Heunen and M. Karvonen. Limits in dagger categories. *Theory and Applications of Categories*, 34(18):468–513, 2019.

[11] C. Heunen and M. Karvonen. Monads on dagger categories. *Theory and Applications of Categories*, 31(35):1016–1043, 2016.

Conference Proceedings

[12] M. Karvonen, R. Piedeleu, and F. Zanasi. Towards synthetic cryptography: a string diagrammatic approach. In P. R. D'Argenio and A. Sokolova, editors, *Theoretical Aspects of Computing - ICTAC 2026*, pages 481–491. Springer Nature Switzerland, 2026.

[13] A. Broadbent and M. Karvonen. Categorical composable cryptography. In *Foundations of Software Science and Computation Structures (FoSSaCS)*, volume 13242 of *Lecture Notes in Computer Science*, pages 161–183. Springer, 2022.

[14] S. Abramsky, R. S. Barbosa, M. Karvonen, and S. Mansfield. A comonadic view of simulation and quantum resources. In *2019 34th Annual ACM/IEEE Symposium on Logic in Computer Science (LICS)*, pages 1–12. IEEE, 2019.

[15] C. Heunen, R. Kaarsgaard, and M. Karvonen. Reversible effects as inverse arrows. In *Mathematical Foundations of Programming Semantics (MFPS)*, volume 341 of *Electronic Notes in Theoretical Computer Science*, pages 179–199. Elsevier, 2018.

[16] M. Karvonen. Categories of empirical models. In *Proceedings of QPL 2018*, volume 287 of *Electronic Proceedings in Theoretical Computer Science*, pages 239–252, 2018.

[17] C. Heunen and M. Karvonen. Reversible monadic computing. In *Mathematical Foundations of Programming Semantics (MFPS)*, volume 319 of *Electronic Notes in Theoretical Computer Science*, pages 217–237, 2015.

Peer-Reviewed Book Chapters

[18] R. S. Barbosa, M. Karvonen, and S. Mansfield. Closing Bell: Boxing Black Box Simulations in the Resource Theory of Contextuality. In *Samson Abramsky on Logic and Structure in Computer Science and Beyond*, Outstanding Contributions to Logic. Springer, 2023.